MILITARY SPECIFICATION SHEET

BIT, SCREWDRIVER, INSERT TYPE, TORQ-SET, RIBBED, OFFSET CRUCIFORM, EXTERNAL HEXAGON SHANK, MANUALLY DRIVEN

This specification is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the bits described herein shall consist of this specification sheet and the latest issue of MIL-B-85643.

1. REQUIREMENTS:

1.1 Material: UNS T41901 (AISI S-1) or UNS T41902 (AISI S-2) tool steel in accordance with QQ-T-570, Type S1 or S2.

1.2 Dimensions and configuration:

Bit and retaining groove: In accordance with Figure 1 and Table I.

Driver point: In accordance with MS14191(AS).

1.2.1 Offset cruciform driver bit gage: In accordance with Figure 2 and Table II.

1.3 Hardness: Rockwell 58 to 62 HRC.

1.4 Identification: Stamp military part number (see 1.8) and manufacturer's name, trademark or symbols.

1.5 Test recess: The test recess configuration and dimensions shall conform to Figure 3 and Table III.

1.6 Minimum ultimate torque loads: In accordance with Table I.

1.7 Torque fatigue loads: In accordance with Table I.

1.8 Example of driver part number:

M85643/5 - 03

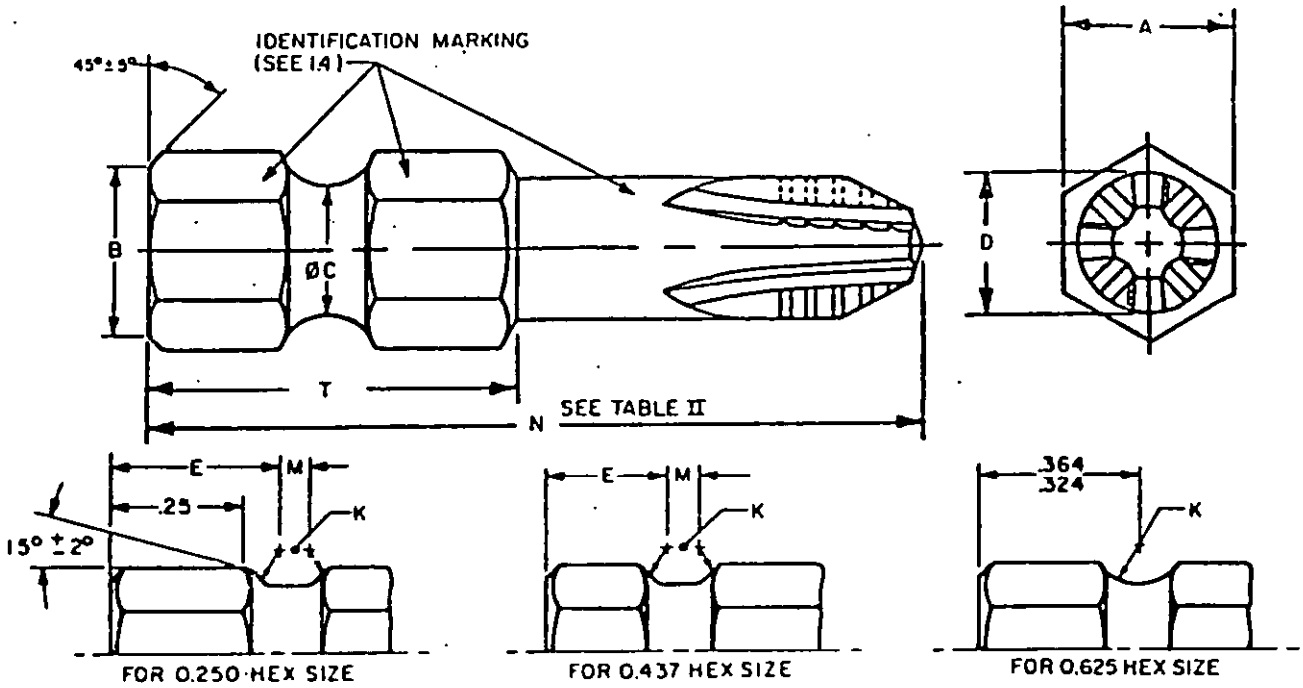
03 — Driver size number (see Table I)

M85643/5 — Basic specification sheet number

AMSC N/A FSC 5120

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DETAIL B - RETAINING GROOVE

FIGURE 1. Bit configuration.

TABLE I. Bit sizes and dimensions.

Driver Size Number	Recess Size	A +.000 -.004	ØB Ref	ØC ±.010	ØD	E ±.020	K Rad	M ±.020	N ±.12	Min Ult Torque 1/	Fatigue Torque 1/
-06	6	.250	.203	.188	.185/.180	.375	.094	.062	1.25	35	16
-08	8				.247 .242					65	30
-03	10									100	45
-04	1/4	.437	.391	.344	.434 .429	.250	.109	.047		250	115
-05	5/16									500	200

1/ Units are (lbf-In.)

Dimensions are in inches.

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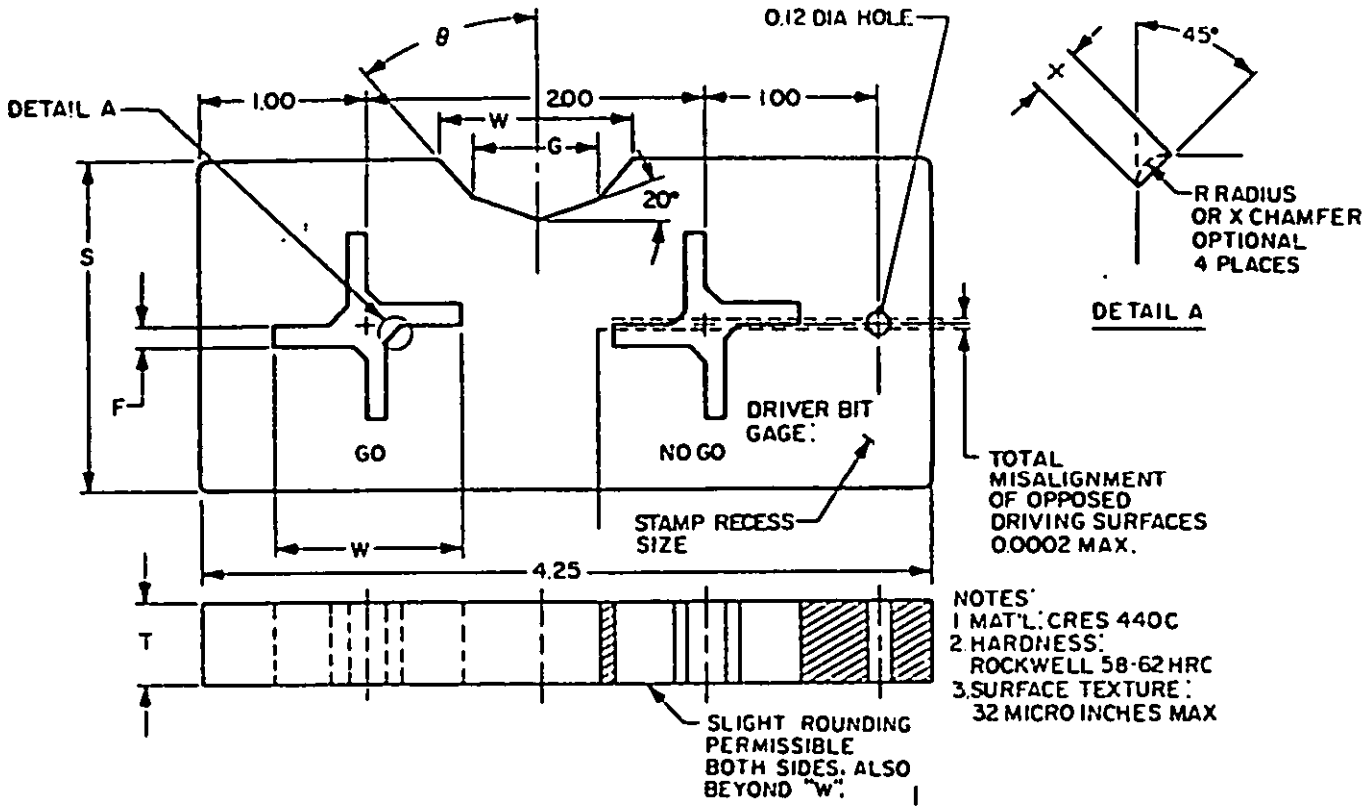


FIGURE 2. Driver bit gage.

TABLE II. Dimensions for offset cruciform driver bit gage.

SIZE	S MIN (WIDTH)	T +.005 (THICK)	GO GAGE DIMENSIONS						NO-GO GAGE DIMENSIONS					
			W ±.005	G ±.005	F +.0002 -.0000	R +.000 -.001	X +.000 -.001	+1 -0°	W ±.005	G ±.005	F +.0000 -.0002	R +.000 -.001	X +.000 -.001	+1° -0°
6	1.00	.187	.188	.083	.0255	.018	.026	47	.188	.083	.0250	.018	.026	47
8	1.00	.187	.223	.097	.0305	.018	.026	47	.223	.097	.0300	.018	.026	47
10	1.00	.187	.258	.112	.0355	.025	.036	47	.258	.112	.0350	.025	.036	47
1/4	1.00	.187	.340	.148	.0475	.025	.036	48	.340	.148	.0455	.025	.036	48
5/16	1.00	.187	.372	.186	.0605	.031	.044	48	.372	.186	.0585	.031	.044	48

ALL DIMENSIONS IN INCHES.
TOLERANCE: ±0.015 UNLESS OTHERWISE NOTED.

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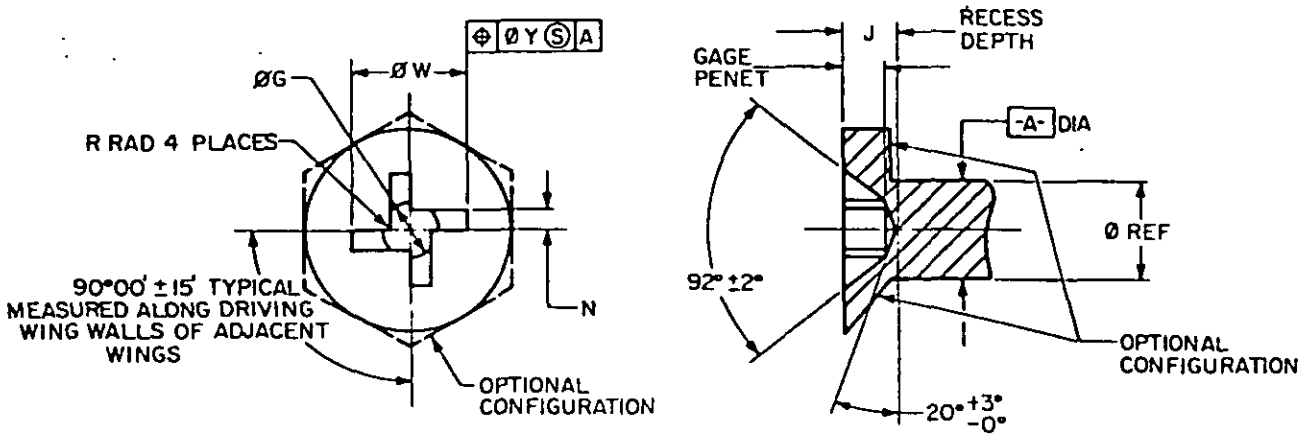


FIGURE 3. Test recess.

TABLE III. Dimensions for test recess with restricted gage penetration for testing offset cruciform bits.

NOMINAL RECESS SIZE	DIMENSIONS									
	N		G DIA		R RAD		Y DIA	/W WING DIA	J RECESS DEPTH MAX REF	GAGE PENET
	MAX	MIN	MAX	MIN	MAX	MIN	TOL			
6	.029	.026	.082	.078	.030	.016	.012	.182 .172	.066	.0500 .0480
8	.034	.031	.096	.092				.215 .205	.078	.0595 .0575
10	.039	.036	.111	.107	.040	.023		.248 .238	.090	.0685 .0665
1/4	.051	.048	.147	.143			.016	.325 .315	.118	.0890 .0870
5/16	.064	.061	.184	.180	.052	.031		.357 .347	.122	.0860 .0840

NOTES:

1. DIMENSIONS ARE IN INCHES.
2. MATERIAL: UNS K74015 OR UNS T20811 ALLOY STEEL PER AMS 6485 OR AMS 6487.
3. HEAT TREATMENT: 220,000 PSI MINIMUM TENSILE STRENGTH. ROCKWELL 46-49HRC.
4. FINISH: CADMIUM PLATE PER QQ-P-416, TYPE II, CLASS 2.
5. RECESSES TO BE GAGED PER MS14191 EXCEPT GAGE PENETRATION IS LIMITED TO THIS DRAWING.

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1.9 ACR and Torq-Set are registered trademarks of Phillips Screw Company. Patent 4,187,892 expires 12 February 1997. The Government does not have a royalty-free license.

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